Authors' Responses

WORKING SKELETON: NOT FOR SUBMISSION

Journal Title:	IEEE Robotics and Automation Letters
Reference Number:	26-3841v2
Title of Paper:	VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation
Authors:	Anonymous Authors

We sincerely thank the Associate Editor and the two anonymous reviewers for their careful assessment of the revised manuscript. We appreciate their recognition of the improvements made in the previous revision and their clear identification of the remaining issues.

[Replace this paragraph with a concise summary of the completed revision after every response below has been factually verified. The summary should cover teacher-reward reproducibility, baseline fairness, result consistency, statistical analysis, the scope of the VLM contribution, and the strengthened limitations statement.]

The detailed point-by-point responses follow. Reviewer comments are reproduced in bold italic text. All red bracketed text is an internal drafting prompt and must be replaced before submission.

RESPONSE TO THE ASSOCIATE EDITOR

We thank the Associate Editor for summarizing the remaining concerns and for directing us to the detailed reviews.

Associate Editor Comment. ***The revised manuscript has improved considerably, and the reviewers acknowledge the authors' efforts in addressing the previous comments. However, several important concerns remain, particularly regarding the RL teacher-training procedure, the fairness of the baseline comparison, inconsistencies in the reported results, and the evidence supporting the contribution of VLM-guided adaptation. These issues need to be addressed to properly support the claimed contributions. Please see the individual reviews for further details.***

[Response:] Drafting checklist:

- **[Summarize the exact teacher-reward equations, weights, scales, and truncation/bootstrap treatment added to the paper or supplement.]**
- **[State the verified resolution of the DexTrAH/FABRICS comparison: either report a fair re-tuned baseline or withdraw the unsupported superiority claim and relabel the transferred implementation accurately.]**
- **[Summarize the corrected Figure 3/Table III relationship and the unified definition of the reported means.]**
- **[Summarize the final evidence and claim scope for VLM-guided adaptation without exceeding what the audited experiments establish.]**

RESPONSE TO REVIEWER #29

We sincerely thank the reviewer for recognizing the substantial improvements in the revised manuscript and for identifying the remaining interpretive, statistical, and reproducibility issues. Below are our point-by-point responses.

Comment 1. ***The revision appropriately removes formal "safety-critical" claims. Please maintain this discipline throughout the manuscript and consistently describe the method as empirical risk reduction under task-specific unsafe-event definitions.***

[Response:] Thank you for recognizing this clarification. We maintain the interpretation of safety as empirical reduction in the four task-specific unsafe-event categories defined in our study. As stated in the Methods section, teacher takeover is empirically safer rather than an absolute safety guarantee. Our framework does not provide formal safety or constraint-satisfaction guarantees.

Comment 2. ***Real-world unsafe-event rates on unseen objects remain around 39%, which is substantial. This limitation should be emphasized prominently in the abstract, discussion, and conclusion to avoid implying deployment-ready safety.***

[Response:] Thank you for emphasizing this limitation. The revised abstract now states:

> "Meanwhile, the distilled student policy achieves comparable lift success rates and demonstrates successful transfer from simulation to the real robot. Generalization to unseen objects remains a clear limitation, with a 39.2% unsafe-event rate in real-world trials."

The discussion in the Real-World Experimental Results subsection already explicitly describes the limitation on unseen objects and its likely causes. We corrected the pooled seen-object rates in this passage to match the verified trial counts.

> "Unseen objects show a clear generalization gap: pooled lift success drops from 83.6% to 64.6%, while the unsafe rate rises from 14.8% to 39.2%, mainly due to more harmful collisions (4.8% to 18.8%) and object escapes (8.1% to 15.8%). Among the six representative objects shown, no palm-flip or hand-too-far events are observed, suggesting that their real-world failures are dominated by contact-rich interactions rather than gross pose instabilities. Overall, the results indicate that the remaining sim-to-real gap on unseen geometries is largely attributable to a contact-strategy mismatch, i.e., the policy fails to adapt its contact interactions to novel object shapes."

The Conclusion identifies the contact-strategy mismatch as a priority for improving generalization to unseen objects.

> "Future work should develop stronger teachers through safety-aware training and improve generalization to unseen objects by addressing the contact-strategy mismatch between simulation and hardware."

Comment 3. ***The real-robot difference between VLM-guided and fixed-threshold critic policies appears modest. Report confidence intervals and paired or proportion tests to determine whether the observed unsafe-rate reductions are statistically distinguishable.***

[Response:] Thank you for pointing out that the real-robot differences are modest. We added 95% confidence intervals and two-sided two-proportion z tests in Appendix Table 8 (Section C.4), reproduced below. Because object poses were randomized independently for each trial, we used an unpaired comparison and pooled trials separately for seen and unseen objects, treating episodes as independent.

Compared with the fixed-threshold policy (c), the VLM-guided policy (d) had an observed unsafe-episode rate 1.43 percentage points lower on seen objects (95% CI $[-3.46, 6.32]$, $p = 0.567$) and 2.50 percentage points lower on unseen objects (95% CI $[-6.28, 11.28]$, $p = 0.577$). Both confidence intervals include zero. We revised the Results to describe these as lower *observed* unsafe-event rates and to make clear that the evidence for a reliable reduction on the real robot remains inconclusive.

Table 8: Real-world comparison of unsafe-event rates. Counts are shown as unsafe episodes over total episodes, followed by the rate. Positive reductions favor ablation (d). Differences use unrounded rates. The abbreviation pp denotes percentage points.

Object group	Ablation (c) unsafe episodes	Ablation (d) unsafe episodes	Observed reduction (pp)	95% confidence interval (pp)	p-value
Seen	68/420 (16.2%)	62/420 (14.8%)	1.43	$[-3.46, 6.32]$	0.567
Unseen	100/240 (41.7%)	94/240 (39.2%)	2.50	$[-6.28, 11.28]$	0.577

Comment 4. ***The Bellman critic achieves a 5.8% false-negative rate on unsafe episodes. Please discuss the consequences of these missed warnings and whether threshold adjustment can trade task success for lower false-negative risk.***

[Response:] Thank you for pointing this out. To address your questions, we added the following discussion in Section IV-E (*Discussion and Limitations*).

> "In passive evaluation, the Bellman critic achieves a 5.8% false-negative rate on unsafe episodes, and during distillation the action disagreement branch can still trigger a teacher takeover on its own when the critic misses a warning.
>
> A higher threshold can reduce missed warnings. Extra takeovers can interrupt attempts that would have succeeded, and they can also rescue attempts that would have failed."

[TODO: Check whether the available data support a threshold sweep and report the results or explain the evidence gap.]

Comment 5. ***Clarify whether critic metrics are computed at the episode or time-step level and how early an unsafe event is detected before termination. Warning lead time is important for intervention usefulness.***

[Response:] Thank you for pointing out this question. At each control step, the Bellman critic scores the current state and the student's proposed action. In the passive evaluation, we count an unsafe episode as detected if the score falls below the critic threshold at least once before termination. The false-negative rate is the fraction of unsafe episodes with no such warning. Task-outcome prediction accuracy is also reported per episode.

Regarding warning lead time, we define it as the elapsed time between the first critic warning and unsafe termination. If the critic warns more than once, we use the first warning because it marks the earliest opportunity for teacher takeover. In post-training evaluations of the student from ablation (d), episodes were capped at 5 s and ran at a control frequency of 60 Hz. The mean warning lead time was 86 control steps, or approximately 1.43 s.

Comment 6. ***The teacher policy is implicitly treated as a safe fallback, yet teacher interventions can still generate unsafe contacts. Report teacher unsafe-event rates and explain what happens when teacher and critic recommendations conflict.***

[Response:] Thank you for raising this concern. The teachers had a mean unsafe-episode rate of 11.4% across all 40 training objects. We reproduce Table II from the main manuscript below.

Table II. Teacher policy performance on four illustrative objects from the 40-object training set (200 evaluation episodes per object). Mean is computed over all 40 objects.

Metric (%)	WindCart	Boot	ToolBox	Cow	**Mean**
Lift Success	93.5	100.0	88.5	87.0	91.7
Unsafe Rate Avg	6.5	13.5	13.5	13.5	11.4
Harmful collision	0.0	0.0	6.5	0.0	2.1
Object escape	6.5	6.5	6.5	13.5	8.0
Palm flipped	0.0	6.5	0.5	0.0	1.2
Hand too far	0.0	0.5	0.0	0.0	0.1

The teacher policy supplies a candidate action, and the Bellman critic evaluates the student's proposed action. The teacher provides no separate safety assessment. The intervention gate executes the teacher's action if the critic's predicted success for the student's action falls below its threshold or if teacher–student action disagreement exceeds its threshold. Either condition can trigger takeover independently. After takeover, safety depends on the teacher policy's capabilities, and unsafe events can still occur.

Comment 7. ***Provide the final exact VLM identifier, prompt, decoding settings, visual-sample selection rules, update interval, latency distribution, failure handling, and cost in the main paper or permanent supplementary material.***

[Response:] Thank you for this suggestion. Given the main manuscript's page limit, we will provide the detailed VLM configuration and inference procedure in Section B of the appendix, *VLM Prompt and Inference Details*.

Comment 8. ***The ablations convincingly motivate visual-semantic adaptation on representative training objects. Please state more explicitly whether the VLM adaptation itself was evaluated on novel scene semantics, rather than only novel object geometries.***

[Response:] Thank you for raising this distinction. We evaluated VLM-guided threshold adaptation during simulated distillation on training objects under randomized visual conditions. We did not separately evaluate the VLM on novel scene semantics, such as new tasks or hazard types. The unseen-object experiments in Table IV evaluate the frozen student policy's generalization to new object geometries on the real robot, where the VLM is inactive.

Comment 9. ***The current unsafe-event taxonomy is task-specific. Clarify which events are directly sensor-observable on hardware and which depend on privileged simulation signals during training or evaluation.***

[Response:] All four unsafe-event categories can be identified using sensors on the physical robot. Harmful collisions can be detected by estimating fingertip contact forces from the InspireHand's motor-current readings. Palm flips can be detected by inferring palm orientation from joint-position measurements. Object escape and hand-too-far events can be detected by tracking the relative positions of the hand and object with an RGB-D camera.

Comment 10. ***Please add a compact limitations paragraph covering the remaining sim-to-real contact-strategy mismatch, external VLM dependence, task-specific safety definitions, and absence of formal constraint guarantees. No major methodological restructuring appears necessary.***

[Response:] Thank you for this suggestion. We added a limitations paragraph in Section IV-E (*Discussion and Limitations*), immediately before the conclusion. It discusses the remaining sim-to-real contact-strategy mismatch, explicitly retaining the 39.2% unsafe-event rate on unseen objects; dependence on an external proprietary VLM; the task-specific scope of the unsafe-event definitions; and the absence of formal safety or constraint-satisfaction guarantees. It also clarifies that the VLM, teacher, and critic are used during simulation distillation rather than as runtime safety monitors on the physical robot.

[TODO: Attach a screenshot of the limitations paragraph in Section IV-E from the highlighted manuscript.]

RESPONSE TO REVIEWER #31

We sincerely thank the reviewer for the detailed technical assessment, for recognizing the expanded real-robot evaluation and reproducibility material, and for identifying the remaining issues in teacher training, baseline fairness, result consistency, and the scope of the VLM contribution.

Major Comment M1. ***The reward for teacher-policy training is described only verbally. Table I lists only the purpose of each reward term; the exact formulation, weights, and scales of the terms are not given in the manuscript. Since teacher training is the starting point of every subsequent stage, the work does not appear reproducible in its current form. Please provide the exact formulation and weights of each reward term.***

[Response:] We thank the reviewer for identifying this reproducibility gap. We agree that the verbal descriptions in Table I were insufficient to reproduce the teacher-training objective. We have therefore added a new "Teacher-Reward Formulation" subsection to Supplementary Section A.2. We reproduce the added formulation below so that the complete correction is also available directly in this response.

At each control step t, all state and action quantities below are evaluated at that step, and the raw environment reward is

$$r = r_{\text{dist}} + r_{\text{curl}} + r_{\text{align}} + r_{\text{contact}} + r_{\text{near}} + r_{\text{goal}} + r_{\text{lift}} + r_{\text{action}}.$$

Let $\mathbf{p}_{o,t}$ and $\mathbf{p}_g$ denote the object and desired goal positions, respectively. For $\mathscr{H} = \{\text{palm center}, \text{five fingertips}\}$, we define

$$d_t^{\text{hand}} = \max_{i \in \mathscr{H}} \|\mathbf{p}_{i,t} - \mathbf{p}_{o,t}\|_2, \qquad d_t^{\text{goal}} = \|\mathbf{p}_{o,t} - \mathbf{p}_g\|_2,$$
$$d_t^z = |p_{o,t}^z - p_g^z|, \qquad \Delta\mathbf{a}_t = \mathbf{a}_t - \mathbf{a}_{t-1}.$$

For each fingertip $i \in \mathscr{F} = \{\text{index}, \text{middle}, \text{ring}, \text{little}, \text{thumb}\}$, the contact and grasp indicators are

$$b_{i,t} = \mathbb{I}\left[\max_j \|\mathbf{f}_{i,j,t}\|_1 > \varepsilon_f\right], \qquad I_t^{\text{any}} = \mathbb{I}\left[\sum_{i \in \mathscr{F}} b_{i,t} > 0\right], \qquad I_t^{\text{grasp}} = b_{\text{thumb,t}}\,\mathbb{I}\left[\sum_{i \in \mathscr{F}} b_{i,t} \geq 2\right].$$

Here $\mathbf{f}_{i,j,t}$ is a vector in the simulator contact-force matrix for fingertip i; I_t^{any} denotes any fingertip–object contact, whereas I_t^{grasp} requires the thumb and at least one other fingertip. With n_t denoting the elapsed environment steps, the eight sub-rewards are

$$\begin{aligned}
r_{\text{dist}} &= w_{\text{dist}} \exp(-\kappa_{\text{dist}} d_t^{\text{hand}}), \\
r_{\text{curl}} &= \text{clip}\left(-w_{\text{curl}} \|\mathbf{q}_t^H - \mathbf{q}_{\text{curl}}\|_2^2, -c_{\text{curl}}, 0\right), \\
r_{\text{align}} &= -w_{\text{align}} \left[\arccos\Big(\text{clip}((\mathbf{d}_t^{\text{palm}})^\top \mathbf{e}_{-z}, -1, 1)\Big)\right]^2, \\
r_{\text{contact}} &= w_{\text{contact}} I_t^{\text{grasp}}, \\
r_{\text{near}} &= w_{\text{near}}\, n_t\, \mathbb{I}[d_t^{\text{hand}} < \varepsilon_{\text{near}}], \\
r_{\text{goal}} &= w_{\text{goal}} \exp(-\alpha_t d_t^{\text{goal}}) I_t^{\text{any}}, \\
r_{\text{lift}} &= w_{\text{lift}} \exp(-\kappa_{\text{lift}} d_t^z) I_t^{\text{any}} \mathbb{I}[n_t \geq n_{\text{lift}}], \\
r_{\text{action}} &= -w_{\text{action}} \left(\|\Delta\mathbf{a}_t^{\text{arm}}\|_2^2 + \lambda_H \|\Delta\mathbf{a}_t^{\text{hand}}\|_2^2\right).
\end{aligned}$$

Here $\mathbf{q}_t^H$ contains the actuated finger-joint positions, $\mathbf{q}_{\text{curl}}$ is the desired curling posture, $\mathbf{d}_t^{\text{palm}}$ is the world-frame unit vector along the palm-down axis, $\mathbf{e}_{-z}$ is the world-down unit vector, and

$$\mathbf{a}_t = \begin{bmatrix} \mathbf{a}_t^{\text{arm}} \\ \mathbf{a}_t^{\text{hand}} \end{bmatrix}.$$

Before PPO uses the reward, the sum of all eight terms is multiplied by η_{PPO}, i.e., $r^{\text{PPO}} = \eta_{\text{PPO}} r$. The individual reward terms are not normalized separately. The numerical parameters used in all experiments are reproduced below.

Teacher-reward parameters used in all experiments.

Component	Parameter values
Hand distance	$w_{\text{dist}} = 5$, $\kappa_{\text{dist}} = 10$
Finger curl	$w_{\text{curl}} = 0.1$, $c_{\text{curl}} = 2$, $\mathbf{q}_{\text{curl}} = [0,0,0,0,0.4,0.1]$
Palm alignment	$w_{\text{align}} = 0.1$
Contact bonus	$w_{\text{contact}} = 10$, $\varepsilon_f = 10^{-4}$
Stay-close	$w_{\text{near}} = 0.005$, $\varepsilon_{\text{near}} = 0.2\,\text{m}$
Goal tracking	$w_{\text{goal}} = 8$, $\alpha_0 = 15$, $\Delta\alpha = 0.1$, $m_{\max} = 50$
Lift	$w_{\text{lift}} = 10$, $\kappa_{\text{lift}} = 8.5$, $n_{\text{lift}} = 60$ steps
Action smoothness	$w_{\text{action}} = 0.01$, $\lambda_H = 3$
Global PPO scaling	$\eta_{\text{PPO}} = 0.01$

Finally, automatic domain randomization progressively sharpens the goal-tracking reward while leaving the lift weight fixed:

$$\alpha_t = \alpha_0 + \Delta\alpha\, m_t, \qquad m_t \in \{0, \ldots, m_{\max}\}.$$

We verified every equation and value directly against the implementation and configuration used to train the reported teacher policies. The complete formulation and parameter table can be found in Supplementary Section A.2, "Teacher-Reward Formulation."

Major Comment M2. ***The claimed superiority of the authors' RL recipe is not supported. (1) The authors explain the performance gap between ablations (a) and (b) by stating that the FABRICS controller was tuned for the original platform and transfers poorly to the new hardware. This implies that the DexTrAH baseline was not tuned for the authors' hardware, and therefore the comparison cannot serve as evidence that the authors' recipe produces more effective teacher policies. A comparison against a DexTrAH baseline re-tuned for the authors' platform, or additional experiments and analysis that can demonstrate the superiority of the recipe, is required. (2) The difference between the "DexTrAH-RGB-style RL recipe" and "our RL recipe" is not defined in the manuscript. Please state explicitly how the two recipes differ. (3) The second contribution claims "a safety-aware RL reward design that explicitly accounts for risky events." However, the reward terms in Table I are generic shaped-reward terms, and no term explicitly penalizes the unsafe events defined by the authors, such as collisions or object escape. Early termination does penalize unsafe events implicitly by truncating the future return, but it is indistinguishable from a timeout unless truncation is bootstrapped. The manuscript does not specify any of these. Please either add an ablation of the reward terms or of the termination conditions, or moderate the contribution statement.***

[Response:]

(1) We clarify that the submitted DexTrAH/FABRICS baseline was not an untuned transfer. We adapted and retuned the controller for the Tiangong 2 Pro and InspireHand, including the robot geometry, constraints, action interface, and controller parameters. On our platform, our direct-control teacher recipe achieved at least comparable task performance while avoiding the human-grasp-motion processing used to construct DexTrAH's hand action space: embodiment-specific motion retargeting, PCA action-manifold construction, and the associated FABRICS tuning. The direct normalized motor-command design is inspired by UniDexGrasp++ (Wan et al., ICCV 2023). We therefore make only the scoped implementation claim that comparable teacher performance can be obtained with substantially less embodiment-specific data processing and controller engineering; we do not claim intrinsic

superiority of the underlying RL algorithm. Consistent with that narrower claim, we removed the two FABRICS-based ablations from the primary contribution argument and removed the corresponding research question.

[Add the verified teacher-only evidence here and replace this note with the final numerical statement: the supplementary teacher-only figure should plot evaluation lift-success versus environment steps for the retuned DexTrAH/FABRICS teacher and our direct-control teacher under the same objects, initialization distribution, horizon, PPO budget, and matched random seeds. Report the mean with a 95% confidence interval, the final paired difference across all 40 objects with a pre-specified equivalence margin, and the corresponding unsafe-termination rates. Use "comparable" only if the verified interval supports that conclusion.]

(2) The main implementation difference is the action-control interface. The submitted DexTrAH-inspired baseline retained the retuned FABRICS controller and its human-motion-derived PCA hand-action manifold between the RL policy and the robot joints. Our teacher-training implementation removes FABRICS and maps 13 normalized policy outputs one-to-one to joint-position targets for the seven arm and six hand DoFs. Revised Sec. III-C now states both this difference and the connection to UniDexGrasp++ explicitly. All retained ablations use the same teacher setup and differ only in their student-distillation and intervention mechanisms.

(3) Consistent with the reduced claim, we deleted the following contribution from the previous manuscript:

> *"We propose a safety-aware RL reward design that explicitly accounts for risky events in arm–hand control scenarios, including harmful collisions, object escape, palm flipping, and the hand moving too far from the workspace, thereby encouraging safer exploration during teacher training."*

Unsafe events are handled through early termination rather than through an explicit negative reward term. The timeout horizon is $T_{\max} = 10\,\mathrm{s}$ and was rarely reached in our experiments. Nevertheless, we agree that early termination should not be presented as an explicit unsafe-event reward, and we have therefore removed the corresponding contribution claim. The termination conditions and timeout treatment are documented in Supplementary Section A.3. We retain the exact reward equations, weights, and implementation details in Supplementary Section A.2 solely to make the teacher-policy training procedure reproducible.

Major Comment M3. ***The unsafe rates in Fig. 3 and Table III are inconsistent. In the "Comparison Unsafe Episode Rate" panel of Fig. 3, (c) is the highest, (b) and (e) are nearly indistinguishable, and (f) is the lowest. In contrast, Table III reports (b) 17.4 > (c) 16.7 > (e) 11.9 > (f) 9.3, which differs in both values and ordering. Please revise so that the two results are consistent with each other.***

[Response:] Thank you for highlighting this apparent discrepancy. We checked the raw logs and evaluation data and confirmed that Figure 3 and Table III correctly report their respective measurements. The unsafe-episode-rate curves in Figure 3 show how unsafe events evolve during online distillation, whereas Table III reports standalone evaluation of the resulting student-policy checkpoints after training.

These measurements support two complementary findings: reduced unsafe exploration during early distillation and lower unsafe-event rates for the resulting student policies in post-training evaluation. We do not claim that the endpoints of the training curves must reproduce the numerical values or ordering in Table III, because the two measurements assess different phases under different conditions.

Thus, the figure and table are consistent with the scope of our claims, although their values and rankings need not coincide. We have revised both captions to clarify this distinction.

The revised captions are reproduced below, with changes highlighted in yellow.

Fig. 3: **Ablation studies.** The first three panels compare four distillation variants in terms of student lift success, teacher intervention rate β, and unsafe-episode rate, respectively, as they evolve during online distillation . The rightmost panel shows the VLM-guided trajectory of the intervention operating point . Shaded regions indicate variability across runs (mean $\pm$ std). Curves correspond to vanilla DAgger (blue), vanilla SafeDAgger (orange), DexSafeDagger (green), and VLM-DexSafeDagger (red).

TABLE III: Performance of ablations (a)–(f) on four illustrative objects from the 40-object training set (200 post-training evaluation episodes per object). Each ablation reports lift success and average unsafe rate. Mean is computed over all 40 objects.

Major Comment M4. ***The contribution of the VLM is not established by the (e) vs. (f) comparison alone. In particular, the third contribution claims effectiveness "across different visual contexts and unsafe scenarios," but the results currently presented do not clearly verify this claim. Please either moderate the corresponding contribution statement or provide additional analysis.***

[Response:] Thank you for this comment. We agree that the comparison between fixed-threshold DexSafeDagger and VLM-DexSafeDagger alone does not isolate the contribution of visual input. Our analysis therefore also includes a non-VLM adaptive curriculum and an LLM-only variant, as described in the Ablation Studies subsection and reported in Table III.

Using the revised manuscript's ablation labels, the comparison forms the following ladder:

- **(c) DexSafeDagger:** fixed intervention thresholds based on teacher–student disagreement and the Bellman success-value critic.
- **(f) Adaptive curriculum:** thresholds adapted using recent lift-success and unsafe-episode statistics, without a language or vision-language model.
- **(e) LLM-DexSafeDagger:** language-model-guided adaptation using rollout statistics without visual observations.
- **(d) VLM-DexSafeDagger:** adaptation using rollout statistics together with visual observations.

These complementary comparisons assess the proposed VLM-guided strategy against both fixed thresholds and alternative adaptive strategies. In particular, the LLM-only variant matches the VLM variant except that it receives no images. The comparison between (e) and (d) therefore addresses the contribution of visual input beyond adaptation based on rollout statistics alone. Our argument for the VLM contribution relies on this broader ablation analysis, rather than solely on the fixed-threshold versus VLM comparison cited by the reviewer, which corresponds to (c) versus (d) in the revised manuscript.

Regarding "different visual contexts," we refer to the randomized visual conditions in our simulation setup: lighting intensity and direction, background textures, object materials, and table appearance. We use DexTrAH-RGB-style domain randomization during both student distillation and the post-training evaluation reported in Table III. Consequently, the results in Table III aggregate performance over these varied visual conditions, rather than a single fixed visual setting. Together with the ablation ladder above, these results support the aggregate effectiveness of VLM-guided adaptation within the evaluated randomized setting. This is the scope intended by "different visual contexts."

By "different unsafe scenarios," we refer to the four task-specific unsafe-event categories defined in our study: harmful collision, object escape, palm flip, and hand-too-far. Figure 4 already decomposes the unsafe-episode rate into these

categories and compares their evolution during online distillation across the four intervention variants for two illustrative objects. The accompanying Results discussion identifies collision and object escape as the dominant failure modes and describes how intervention suppresses their early-training spikes. This analysis complements the aggregate comparisons in Figure 3 and Table III.

If the reviewer still considers the wording of the contribution potentially misleading, we will consider adopting more cautious and precise wording.

Minor Comment m1. ***One paragraph in Sec. III-A has a line spacing that differs from the rest of the text. Please check the formatting of the entire manuscript.***

[Response:] Thank you for bringing this formatting concern to our attention. We carefully checked Section III-A and visually inspected every page of the revised manuscript, including paragraph spacing, captions, tables, equations, and section transitions. We confirmed that body-text line spacing is consistent throughout the revised manuscript, including Section III-A.

Minor Comment m2. ***The "Mean" columns of Table II and Table III appear to average over different populations. Please unify the definition of "Mean" across the two tables. In addition, please briefly state why these four objects were selected from the 40-object set.***

[Response:] We thank the reviewer for identifying the inconsistent aggregation populations. We corrected this issue by recalculating every entry in Table II's "Mean" column directly from the raw evaluation data for all 40 objects, so that Tables II and III now use the same 40-object population. Both table captions now explicitly state that "Mean is computed over all 40 objects."

The four objects were selected only as a compact illustrative subset to show object-level results within the available table space. They were not selected using a statistical, difficulty-based, or performance-based criterion, and no conclusion is based specifically on these four objects. We agree that the word "representative" incorrectly implied a formal selection procedure and have replaced it with "illustrative."

Minor Comment m3. ***Fig. 4 is difficult to read at print quality. Please also explain why only two objects (windcart and cow) are shown.***

[Response:] Thank you for pointing this out. Image compression appears to have reduced the readability of Fig. 4 at print size. To address this, we will enlarge the legend and increase the font sizes to make the figure easier to read.

We show WindCart and Cow as illustrative examples of the failure-mode dynamics during training. Including additional objects would make the plots overly dense and difficult to read at print size. Figure 4 therefore presents a compact illustration, while Table III reports the aggregate quantitative results over all 40 objects.

[Response:] Drafting checklist:

- **[TODO: Regenerate Fig. 4 from the original plotting source at publication quality; enlarge the legend and fonts, avoid quality-reducing image compression, and verify readability at final print size.]**

We thank the Associate Editor and the reviewers again for their constructive feedback. **[Replace this closing with a concise statement of the verified revisions and their effect on the manuscript's claims.]**